角动量算符

（二）角动量算符

（1）角动量算符的形式

经典力学中，若动量为 **p**，相对点**O** 的位置矢量为 **r** 的粒子绕 **O** 点的角动量是：

$$\vec{L} = \vec{r} \times \vec{p}$$

$$\hat{\vec{L}} = \hat{\vec{r}} \times \hat{\vec{p}} = -i\hbar\vec{r} \times \nabla$$

量子力学基本假定

角动量算符

(I) 直角坐标系

$$\hat{\vec{L}} = -i\hbar\vec{r}\times\nabla$$

$$\begin{cases} \hat{L}_x = y\hat{p}_z - z\hat{p}_y = -i\hbar(y\frac{\partial}{\partial z} - z\frac{\partial}{\partial y}) \\ \hat{L}_y = z\hat{p}_x - x\hat{p}_z = -i\hbar(z\frac{\partial}{\partial x} - x\frac{\partial}{\partial z}) \\ \hat{L}_z = x\hat{p}_y - y\hat{p}_x = -i\hbar(x\frac{\partial}{\partial y} - y\frac{\partial}{\partial x}) \end{cases}$$

角动量平方算符

$$\hat{L}^2 = \hat{L}_x^{\,2} + \hat{L}_y^{\,2} + \hat{L}_z^{\,2}$$

角动量算符

考察角动量算符的乘积：

$$\hat{L}_x\hat{L}_y-\hat{L}_y\hat{L}_x=\left(y\hat{p}_z-z\hat{p}_y\right)\left(z\hat{p}_x-x\hat{p}_z\right)-\left(z\hat{p}_x-x\hat{p}_z\right)\left(y\hat{p}_z-z\hat{p}_y\right)$$

$$=y\hat{p}_z z\hat{p}_x-z\hat{p}_y z\hat{p}_x-y\hat{p}_z x\hat{p}_z+z\hat{p}_y x\hat{p}_z$$

$$-z\hat{p}_x y\hat{p}_z+x\hat{p}_z y\hat{p}_z+z\hat{p}_x z\hat{p}_y-x\hat{p}_z z\hat{p}_y$$

不同维度的位置算符与动量算符的积可交换

$$=y\hat{p}_z z\hat{p}_x-z\hat{p}_x z\hat{p}_y-x\hat{p}_z y\hat{p}_z+z\hat{p}_y x\hat{p}_z$$

$$-z\hat{p}_x y\hat{p}_z+x\hat{p}_z y\hat{p}_z+z\hat{p}_x z\hat{p}_y-x\hat{p}_z z\hat{p}_y$$

$\hat{p}_z z$ 不可乘积交换

$$=y\hat{p}_z z\hat{p}_x+z\hat{p}_y x\hat{p}_z-z\hat{p}_x y\hat{p}_z-x\hat{p}_z z\hat{p}_y$$

$$=y\hat{p}_x\hat{p}_z z+x\hat{p}_y z\hat{p}_z-y\hat{p}_x z\hat{p}_z-x\hat{p}_y\hat{p}_z z$$

$$=y\hat{p}_x\left(\hat{p}_z z-z\hat{p}_z\right)+x\hat{p}_y\left(z\hat{p}_z-\hat{p}_z z\right)$$

$z\hat{p}_z-\hat{p}_z z=i\hbar$

$$=-i\hbar y\hat{p}_x+i\hbar x\hat{p}_y=i\hbar\left(x\hat{p}_y-y\hat{p}_x\right)$$

$\hat{L}_z=x\hat{p}_y-y\hat{p}_x$

$$=i\hbar\hat{L}_z$$

同理

$$\hat{L}_y\hat{L}_z-\hat{L}_z\hat{L}_y=i\hbar\hat{L}_x$$

$$\hat{L}_z\hat{L}_x-\hat{L}_x\hat{L}_z=i\hbar\hat{L}_y$$

角动量算符

考察角动量平方算符和角动量算符的乘积：

$$\hat{L}^2\hat{L}_x-\hat{L}_x\hat{L}^2=\left(\hat{L}_x^{\,2}+\hat{L}_y^{\,2}+\hat{L}_z^{\,2}\right)\hat{L}_x-\hat{L}_x\left(\hat{L}_x^{\,2}+\hat{L}_y^{\,2}+\hat{L}_z^{\,2}\right)$$

$$=\hat{L}_y^{\,2}\hat{L}_x+\hat{L}_z^{\,2}\hat{L}_x-\hat{L}_x\hat{L}_y^{\,2}-\hat{L}_x\hat{L}_z^{\,2}$$

$$\hat{L}_x\hat{L}_y-\hat{L}_y\hat{L}_x=i\hbar\hat{L}_z$$
$$\hat{L}_z\hat{L}_x-\hat{L}_x\hat{L}_z=i\hbar\hat{L}_y$$

$$=\hat{L}_y^{\,2}\hat{L}_x+\hat{L}_z^{\,2}\hat{L}_x-\left(i\hbar\hat{L}_z+\hat{L}_y\hat{L}_x\right)\hat{L}_y-\left(\hat{L}_z\hat{L}_x-i\hbar\hat{L}_y\right)\hat{L}_z$$

整理

$$=\hat{L}_y^{\,2}\hat{L}_x+\hat{L}_z^{\,2}\hat{L}_x-\left(i\hbar\hat{L}_z\hat{L}_y+\hat{L}_y\hat{L}_x\hat{L}_y\right)-\left(\hat{L}_z\hat{L}_x\hat{L}_z-i\hbar\hat{L}_y\hat{L}_z\right)$$

$$=\hat{L}_y\left(\hat{L}_y\hat{L}_x-\hat{L}_x\hat{L}_y\right)+\hat{L}_z\left(\hat{L}_z\hat{L}_x-\hat{L}_x\hat{L}_z\right)-i\hbar L_z\hat{L}_y+i\hbar L_y\hat{L}_z$$

$$\hat{L}_x\hat{L}_y-\hat{L}_y\hat{L}_x=i\hbar\hat{L}_z$$
$$\hat{L}_z\hat{L}_x-\hat{L}_x\hat{L}_z=i\hbar\hat{L}_y$$

$$=\hat{L}_y\left(-i\hbar\hat{L}_z\right)+\hat{L}_z\left(i\hbar\hat{L}_y\right)-i\hbar\hat{L}_z\hat{L}_y+i\hbar\hat{L}_y\hat{L}_z=0$$

同理　$\hat{L}^2\hat{L}_y-\hat{L}_y\hat{L}^2=0$　$\hat{L}^2\hat{L}_z-\hat{L}_z\hat{L}^2=0$

角动量算符

角动量算符乘积满足的关系

$$\hat{L}_x\hat{L}_y-\hat{L}_y\hat{L}_x=i\hbar\hat{L}_z$$

$$\hat{L}_y\hat{L}_z-\hat{L}_z\hat{L}_y=i\hbar\hat{L}_x$$

$$\hat{L}_z\hat{L}_x-\hat{L}_x\hat{L}_z=i\hbar\hat{L}_y$$

角动量平方算符与角动量算符乘积可交换

$$\hat{L}^2\hat{L}_x-\hat{L}_x\hat{L}^2=0$$

$$\hat{L}^2\hat{L}_y-\hat{L}_y\hat{L}^2=0$$

$$\hat{L}^2\hat{L}_z-\hat{L}_z\hat{L}^2=0$$

角动量算符

考察角动量算符的厄密共轭算符

$$\hat{L}_x^{+} = \left(y\hat{p}_z - z\hat{p}_y\right)^{+}$$

$$= \left(y\hat{p}_z\right)^{+} - \left(z\hat{p}_y\right)^{+}$$

利用 $(\hat{O}\hat{A})^{+} = \hat{A}^{+}\hat{O}^{+}$

$$= \hat{p}_z^{+}y^{+} - \hat{p}_y^{+}z^{+}$$

位置、动量算符的厄密共轭算符为自身

$$= \hat{p}_z y - \hat{p}_y z$$

不同维度的位置算符与动量算符的积可交换

$$= y\hat{p}_z - z\hat{p}_y = \hat{L}_x$$

同理可证 $\hat{L}_y^{+} = \hat{L}_y, \hat{L}_z^{+} = \hat{L}_z$

角动量算符的厄密共轭算符是其自身

厄密算符

角动量算符

考察角动量平方算符的厄密共轭算符

$$\hat{L}^2 = \hat{L}_x^{\,2} + \hat{L}_y^{\,2} + \hat{L}_z^{\,2}$$

$$\left(\hat{L}^2\right)^+ = \left(\hat{L}_x^{\,2} + \hat{L}_y^{\,2} + \hat{L}_z^{\,2}\right)^+$$

$$= \left(\hat{L}_x^{\,2}\right)^+ + \left(\hat{L}_y^{\,2}\right)^+ + \left(\hat{L}_z^{\,2}\right)^+$$

利用 $(\hat{O}\hat{A})^+ = \hat{A}^+\hat{O}^+$

$$= \hat{L}_x^{\,+}\hat{L}_x^{\,+} + \hat{L}_y^{\,+}\hat{L}_y^{\,+} + \hat{L}_z^{\,+}\hat{L}_z^{\,+}$$

角动量算符的厄密共轭算符为自身

$$= \hat{L}_x\hat{L}_x + \hat{L}_y\hat{L}_y + \hat{L}_z\hat{L}_z$$

$$= \hat{L}^2$$

角动量平方算符的厄密共轭算符是其自身 厄密算符

角动量算符

考察角动量平方算符的本征态

$$\hat{L}^2 = \hat{L}_x^{\,2} + \hat{L}_y^{\,2} + \hat{L}_z^{\,2}$$

$$= (y\hat{p}_z - z\hat{p}_y)^2 + (z\hat{p}_x - x\hat{p}_z)^2 + (x\hat{p}_y - y\hat{p}_x)^2$$

$$= -\hbar^2[(y\frac{\partial}{\partial z} - z\frac{\partial}{\partial y})^2 + (z\frac{\partial}{\partial x} - x\frac{\partial}{\partial z})^2 + (x\frac{\partial}{\partial y} - y\frac{\partial}{\partial x})^2]$$

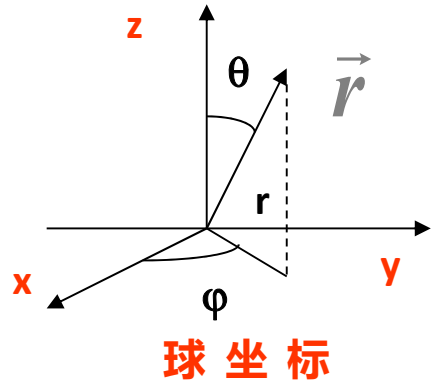
由于角动量平方算符中含有关于 **x**，**y**，**z** 偏导数的交叉项，所以直角坐标下角动量平方算符的本征方程不能分离变量，难于求解，为此我们采用球坐标较为方便。

角动量算符

（II）球坐标

直角坐标与球坐标之间的变换关系

$$\begin{cases} x = r\sin\theta\cos\varphi \\ y = r\sin\theta\sin\varphi \\ z = r\cos\theta \end{cases} \qquad \begin{cases} r^2 = x^2 + y^2 + z^2 & (1) \\ \cos\theta = z/r & (2) \\ \tan\varphi = y/x & (3) \end{cases}$$



对于任意函数 f(r, θ, φ)（其中r, θ, φ都是x, y, z 的函数）则有：

$$\frac{\partial f}{\partial x_i} = \frac{\partial f}{\partial r}\frac{\partial r}{\partial x_i} + \frac{\partial f}{\partial \theta}\frac{\partial \theta}{\partial x_i} + \frac{\partial f}{\partial \varphi}\frac{\partial \varphi}{\partial x_i} \qquad \text{其中} x_1, x_2, x_3 = x, y, z$$

或

$$\begin{cases} \dfrac{\partial}{\partial x} = \dfrac{\partial}{\partial r}\dfrac{\partial r}{\partial x} + \dfrac{\partial}{\partial \theta}\dfrac{\partial \theta}{\partial x} + \dfrac{\partial}{\partial \varphi}\dfrac{\partial \varphi}{\partial x} \\ \dfrac{\partial}{\partial y} = \dfrac{\partial}{\partial r}\dfrac{\partial r}{\partial y} + \dfrac{\partial}{\partial \theta}\dfrac{\partial \theta}{\partial y} + \dfrac{\partial}{\partial \varphi}\dfrac{\partial \varphi}{\partial y} \\ \dfrac{\partial}{\partial z} = \dfrac{\partial}{\partial r}\dfrac{\partial r}{\partial z} + \dfrac{\partial}{\partial \theta}\dfrac{\partial \theta}{\partial z} + \dfrac{\partial}{\partial \varphi}\dfrac{\partial \varphi}{\partial z} \end{cases}$$

角动量算符

$$\begin{cases} r^2 = x^2 + y^2 + z^2 & (1) \\ \cos\theta = z / r & (2) \\ \tan\varphi = y / x & (3) \end{cases} \qquad \begin{cases} x = r\sin\theta\cos\varphi \\ y = r\sin\theta\sin\varphi \\ z = r\cos\theta \end{cases}$$

将（1）式两边分别对 x y z 求偏导数得：

$$\begin{cases} \dfrac{\partial r}{\partial x} = \sin\theta\cos\varphi \\ \dfrac{\partial r}{\partial y} = \sin\theta\sin\varphi \\ \dfrac{\partial r}{\partial z} = \cos\theta \end{cases}$$

$$\frac{\partial r}{\partial x} = \frac{1}{2}\frac{2x}{\sqrt{x^2+y^2+z^2}} = \frac{r\sin\theta\cos\varphi}{r} = \sin\theta\cos\varphi$$

$$\frac{\partial r}{\partial y} = \frac{1}{2}\frac{2y}{\sqrt{x^2+y^2+z^2}} = \frac{r\sin\theta\sin\varphi}{r} = \sin\theta\sin\varphi$$

$$\frac{\partial r}{\partial z} = \frac{1}{2}\frac{2z}{\sqrt{x^2+y^2+z^2}} = \frac{r\cos\theta}{r} = \cos\theta$$

将（2）式两边分别对 x y z 求偏导数得：

$$\begin{cases} \dfrac{\partial\theta}{\partial x} = \dfrac{1}{r}\cos\theta\cos\varphi \\ \dfrac{\partial\theta}{\partial y} = \dfrac{1}{r}\cos\theta\sin\varphi \\ \dfrac{\partial\theta}{\partial z} = -\dfrac{1}{r}\sin\theta \end{cases}$$

将（3）式两边分别对 x y z 求偏导数得：

$$\begin{cases} \dfrac{\partial\varphi}{\partial x} = -\dfrac{1}{r}\dfrac{\sin\varphi}{\sin\theta} \\ \dfrac{\partial\varphi}{\partial y} = \dfrac{1}{r}\dfrac{\cos\varphi}{\sin\theta} \\ \dfrac{\partial\varphi}{\partial z} = 0 \end{cases}$$

角动量算符

将上面结果代回原式得：

$$\begin{cases} \dfrac{\partial}{\partial x}=\sin\theta\cos\varphi\dfrac{\partial}{\partial r}+\dfrac{1}{r}\cos\theta\cos\varphi\dfrac{\partial}{\partial\theta}-\dfrac{1}{r}\dfrac{\sin\varphi}{\sin\theta}\dfrac{\partial}{\partial\varphi} \\ \dfrac{\partial}{\partial y}=\sin\theta\sin\varphi\dfrac{\partial}{\partial r}+\dfrac{1}{r}\cos\theta\sin\varphi\dfrac{\partial}{\partial\theta}+\dfrac{1}{r}\dfrac{\cos\varphi}{\sin\theta}\dfrac{\partial}{\partial\varphi} \\ \dfrac{\partial}{\partial z}=\cos\theta\dfrac{\partial}{\partial r}-\dfrac{1}{r}\sin\theta\dfrac{\partial}{\partial\theta}+0 \end{cases}$$

则角动量算符在球坐标中的表达式为：

$$\begin{cases} \hat{L}_x=i\hbar\left(\sin\varphi\dfrac{\partial}{\partial\theta}+\cot\theta\cos\varphi\dfrac{\partial}{\partial\varphi}\right) \\ \hat{L}_y=-i\hbar\left(\cos\varphi\dfrac{\partial}{\partial\theta}+\cot\theta\sin\varphi\dfrac{\partial}{\partial\varphi}\right) \\ \hat{L}_z=-i\hbar\dfrac{\partial}{\partial\varphi} \end{cases}$$

$$\hat{L}^2=-\hbar^2\left[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right)+\frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\varphi^2}\right]$$

角动量算符

（2）本征方程

（I）$\hat{L}_z$ 的本征方程

$$\hat{L}_z = -i\hbar\frac{\partial}{\partial\varphi}$$

列方程

$$\hat{L}_z\psi(\varphi) = -i\hbar\frac{d}{d\varphi}\psi(\varphi) = l_z\psi(\varphi)$$

解方程

$$\psi(\varphi) = ce^{\frac{i}{\hbar}l_z\varphi}$$

其中c是积分常数，亦可看成归一化系数。

应用波函数标准条件

I。波函数有限条件，要求 ℓ_z 为实数；

II。波函数单值条件，要求当 φ 转过 2π角回到原位时波函数值相等，即：

$$\psi(\varphi) = \psi(\varphi+2\pi) \quad \rightarrow ce^{\frac{i}{\hbar}l_z\varphi} = ce^{\frac{i}{\hbar}l_z(\varphi+2\pi)} \quad \rightarrow e^{\frac{i}{\hbar}l_z 2\pi} = 1$$

$$\text{于是} \quad \frac{2\pi l_z}{\hbar} = 2\pi m \qquad m = 0,\pm1,\pm2,\cdots$$

$$\rightarrow \qquad l_z = m\hbar \qquad m = 0,\pm1,\pm2,\cdots$$

$$\psi(\varphi) = ce^{im\varphi}$$

角动量算符

求归一化系数

$$\psi(\varphi) = ce^{im\varphi}$$

$$\int_0^{2\pi} |\psi|^2 \, d\varphi = c^2 \int_0^{2\pi} d\varphi \; = 2\pi c^2 = 1 \qquad \rightarrow c = \frac{1}{\sqrt{2\pi}}$$

最后得 $\hat{L}_z$ 的本征函数和本征值：

$$\begin{cases} l_z = m\hbar \\ \psi_m(\varphi) = \dfrac{1}{\sqrt{2\pi}} e^{im\varphi} \end{cases} \qquad m = 0, \pm 1, \pm 2, \cdots$$

正交性

$$\frac{1}{2\pi} \int_0^{2\pi} e^{-im\varphi} e^{in\varphi} d\varphi = 0 \qquad (n \neq m)$$

合记之得正交归一化条件

$$\frac{1}{2\pi} \int_0^{2\pi} e^{-im\varphi} e^{in\varphi} d\varphi = \delta_{mn}$$

角动量算符

(II) $\hat{L}^2$ 的本征值问题

$\hat{L}^2$ 的本征值方程可写为：

$$\hat{L}^2 Y(\theta,\varphi) = \lambda\hbar^2 Y(\theta,\varphi)$$

$$-\hbar^2\left[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}(\sin\theta\frac{\partial}{\partial\theta})+\frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\varphi^2}\right]Y(\theta,\varphi)=\lambda\hbar^2 Y(\theta,\varphi)$$

或：

$$-\left[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}(\sin\theta\frac{\partial}{\partial\theta})+\frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\varphi^2}\right]Y(\theta,\varphi)=\lambda Y(\theta,\varphi)$$

其中 **Y(θ,φ)** 是 $\hat{L}^2$ 属于本征值**λħ²** 的本征函数。此方程就是大家熟悉的球谐函数方程，其求解方法在数学物理方法中已有详细的讲述，得到的结论是：

为使 **Y(θ,φ)** 在θ 变化的整个区域**(0, π)**内都是有限的，
则必须满足： $\lambda = \ell(\ell+1)$, 其中 $\ell = 0, 1, 2, \ldots$

角动量算符

$$\hat{L}^2 Y(\theta,\varphi) = \lambda\hbar^2 Y(\theta,\varphi)$$

该方程的解就是球函数$\mathbf{Y_{lm}(\theta,\varphi)}$，其表达式：

$$Y_{lm}(\theta,\varphi) = (-1)^m N_{lm} P_l^m(\cos\theta) e^{im\varphi} \qquad m = 0,1,2,\cdots,l$$

P_l^m为连带Legendre多项式

$$Y_{lm}(\theta,\varphi) = (-1)^m Y_{l-m}^*(\theta,\varphi) \qquad m = -1,-2,-3,\cdots,-l$$

归一化系数，由归一化条件确定

$$\int_0^{2\pi}\int_0^{\pi} Y_{lm}^*(\theta,\varphi) Y_{lm}(\theta,\varphi)\sin\theta d\theta d\varphi = 1 \qquad N_{lm} = \sqrt{\frac{(l-|m|)!(2l+1)}{4\pi(l+|m|)!}}$$

其正交归一条件为：

$$\int_0^{2\pi}\int_0^{\pi} Y_{lm}^*(\theta,\varphi) Y_{l'm'}(\theta,\varphi)\sin\theta d\theta d\varphi = \delta_{ll'}\delta_{mm'}$$

具体计算请参考有关数学物理方法的书籍，在这里就不作详细介绍了。

角动量算符

本征值的简并度

$$Y_{lm}(\theta,\varphi)=(-1)^m N_{lm}P_l^m(\cos\theta)e^{im\varphi} \qquad m=0,1,2,\cdots,l$$

$$Y_{lm}(\theta,\varphi)=(-1)^m Y_{l-m}^*(\theta,\varphi) \qquad m=-1,-2,-3,\cdots,-l$$

由于量子数 ℓ 表征了角动量的大小，所以称为角量子数；**m** 称为磁量子数。

可知，对应一个 ℓ 值，**m** 取值为 **0, ±1, ±2, ±3, ..., ±ℓ** 共 **(2 ℓ +1)**个值。因此当ℓ 确定后，尚有**(2 ℓ +1)**个磁量子状态不确定。

换言之，对应一个ℓ值有**(2 ℓ +1)**个量子状态，这种现象称为简并，ℓ 的简并度是 **(2 ℓ +1)** 度。

角动量算符

$\hat{L}^2$的本征态

$$Y_{lm}(\theta,\varphi)=(-1)^m N_{lm}P_l^m(\cos\theta)e^{im\varphi} \qquad m=0,1,2,\cdots,l$$

$$Y_{lm}(\theta,\varphi)=(-1)^m Y_{l-m}^*(\theta,\varphi) \qquad m=-1,-2,-3,\cdots,-l$$

$$\hat{L}_z=-i\hbar\frac{\partial}{\partial\varphi}$$

$$\hat{L}_z Y_{lm}(\theta,\varphi)=-i\hbar\frac{\partial}{\partial\varphi}\left[(-1)^m N_{lm}P_l^m(\cos\theta)e^{im\varphi}\right]$$

$$=-i\hbar(-1)^m N_{lm}P_l^m(\cos\theta)\frac{\partial}{\partial\varphi}e^{im\varphi}$$

$$=-i\hbar(-1)^m N_{lm}P_l^m(\cos\theta)(im)e^{im\varphi}$$

$$=m\hbar(-1)^m N_{lm}P_l^m(\cos\theta)e^{im\varphi}=m\hbar Y_{lm}(\theta,\varphi)$$

易证**m<0**的情形也满足 $\hat{L}_z Y_{lm}(\theta,\varphi)=m\hbar Y_{lm}(\theta,\varphi)$

$Y_{lm}(\theta,\varphi)$也是 $\hat{L}_z$ 的本征态 ⇒ **$Y_{lm}(\theta,\varphi)$**是 $\hat{L}^2$，$\hat{L}_z$的共同本征函数

角动量算符

角动量升降阶算符

(I) 定义

$$\begin{cases} \hat{L}_+ = \hat{L}_x + i\hat{L}_y \\ \hat{L}_- = \hat{L}_x - i\hat{L}_y \end{cases}$$

性质 $\hat{L}_+^+ = \hat{L}_-,\ \hat{L}_-^+ = \hat{L}_+$

$$\hat{L}_+^+ = (\hat{L}_x + i\hat{L}_y)^+$$

利用 $\hat{O}^+ = \tilde{\hat{O}}^*$

$$= \hat{L}_x^+ - i\hat{L}_y^+$$

角动量算符的厄密共轭算符为自身

$$= \hat{L}_x - i\hat{L}_y = \hat{L}_-$$

同理易证 $\hat{L}_-^+ = \hat{L}_+$

角动量算符

$$\hat{L}_{\pm}Y_{lm}=\hbar\sqrt{l(l+1)-m(m\pm1)}Y_{l,m\pm1}$$
$$=\hbar\sqrt{(l\mp m)(l\pm m+1)}Y_{l,m\pm1}$$

提前了解

$$\begin{cases}\hat{L}_z\hat{L}_{\pm}=\hat{L}_{\pm}(\hat{L}_z\pm\hbar)\\ \hat{L}^2\hat{L}_{\pm}=\hat{L}_{\pm}\hat{L}^2\\ \hat{L}_+\hat{L}_-=\hat{L}^2-\hat{L}_z^2+\hbar\hat{L}_z\\ \hat{L}_-\hat{L}_+=\hat{L}^2-\hat{L}_z^2-\hbar\hat{L}_z\end{cases}$$

角动量算符

$$\hat{L}_z\hat{L}_\pm = \hat{L}_\pm(\hat{L}_z \pm \hbar)$$

$$\hat{L}_z\hat{L}_\pm = \hat{L}_z\left(\hat{L}_x \pm i\hat{L}_y\right)$$

$$= \hat{L}_z\hat{L}_x \pm i\hat{L}_z\hat{L}_y$$

$$\hat{L}_z\hat{L}_x - \hat{L}_x\hat{L}_z = i\hbar\hat{L}_y$$
$$\hat{L}_y\hat{L}_z - \hat{L}_z\hat{L}_y = i\hbar\hat{L}_x$$

$$= \left(i\hbar\hat{L}_y + \hat{L}_x\hat{L}_z\right) \pm i\left(\hat{L}_y\hat{L}_z - i\hbar\hat{L}_x\right)$$

$$= \left(\hat{L}_x \pm i\hat{L}_y\right)\hat{L}_z + \hbar\left(i\hat{L}_y \pm \hat{L}_x\right)$$

$$= \left(\hat{L}_x \pm i\hat{L}_y\right)\hat{L}_z \pm \hbar\left(\hat{L}_x \pm i\hat{L}_y\right)$$

合并

$$= \left(\hat{L}_x \pm i\hat{L}_y\right)\left(\hat{L}_z \pm \hbar\right)$$

利用升降算符定义

$$= \hat{L}_\pm\left(\hat{L}_z \pm \hbar\right)$$

角动量算符

$$\hat{L}^2\hat{L}_{\pm}=\hat{L}_{\pm}\hat{L}^2$$

$$\hat{L}^2\hat{L}_{\pm}=\hat{L}^2\left(\hat{L}_x\pm i\hat{L}_y\right)$$

$$=\hat{L}^2\hat{L}_x\pm i\hat{L}^2\hat{L}_y$$

$$\hat{L}^2\hat{L}_x-\hat{L}_x\hat{L}^2=0$$

$$\hat{L}^2\hat{L}_y-\hat{L}_y\hat{L}^2=0$$

$$=\hat{L}_x\hat{L}^2\pm i\hat{L}_y\hat{L}^2$$

$$=\left(\hat{L}_x\pm i\hat{L}_y\right)\hat{L}^2$$

$$=\hat{L}_{\pm}\hat{L}^2$$

角动量算符

$$\hat{L}_+\hat{L}_- = \hat{L}^2 - \hat{L}_z^2 + \hbar\hat{L}_z$$

$$\hat{L}_+\hat{L}_- = \left(\hat{L}_x + i\hat{L}_y\right)\left(\hat{L}_x - i\hat{L}_y\right)$$

$$= \hat{L}_x\hat{L}_x - i\hat{L}_x\hat{L}_y + i\hat{L}_y\hat{L}_x + \hat{L}_y\hat{L}_y$$

$$= \hat{L}_x{}^2 + \hat{L}_y{}^2 - i\left(\hat{L}_x\hat{L}_y - \hat{L}_y\hat{L}_x\right)$$

$$\hat{L}_x\hat{L}_y - \hat{L}_y\hat{L}_x = i\hbar\hat{L}_z$$

$$= \hat{L}^2 - \hat{L}_z^2 + \hbar\hat{L}_z$$

同理

$$\hat{L}_-\hat{L}_+ = \hat{L}^2 - \hat{L}_z^2 - \hbar\hat{L}_z$$

角动量算符

证明：

$$\hat{L}_{\pm}Y_{lm}=\hbar\sqrt{l(l+1)-m(m\pm1)}Y_{l,m\pm1}$$

$$=\hbar\sqrt{(l\mp m)(l\pm m+1)}Y_{l,m\pm1}$$

考察$\hat{L}_{\pm}Y_{lm}$这个态被$\hat{L}_z$和$\hat{L}^2$作用后的情况

$\hat{L}_z\left(\hat{L}_+Y_{lm}\right)$

$\hat{L}_z\hat{L}_{\pm}=\hat{L}_{\pm}(\hat{L}_z\pm\hbar)$ $=\hat{L}_+(\hat{L}_z+\hbar)Y_{lm}$

$\hat{L}_zY_{lm}=m\hbar Y_{lm}$ $=\hat{L}_+(m\hbar+\hbar)Y_{lm}$

$=(m+1)\hbar(\hat{L}_+Y_{lm})$

(L_+Y_{lm})是 L_z 的本征函 数
对应本征值为(m+1) $\hbar$

$\hat{L}^2\left(\hat{L}_+Y_{lm}\right)$

$\hat{L}^2\hat{L}_{\pm}=\hat{L}_{\pm}\hat{L}^2$ $=\hat{L}_+\hat{L}^2Y_{lm}$

$\hat{L}^2Y_{lm}=l(l+1)\hbar^2Y_{lm}$ $=\hat{L}_+[l(l+1)\hbar^2]Y_{lm}$

$=l(l+1)\hbar^2(\hat{L}_+Y_{lm})$

(L_+Y_{lm})是 L^2 的本征函 数
对应本征值为l (l+1) $\hbar^2$

由于相应于这些本征值的本征函数是$Y_{l,m+1}$，所以L_+Y_{lm}与$Y_{l,m+1}$仅差一个常数，即

$$\hat{L}_+Y_{lm}=a_{lm}Y_{l,m+1}$$

同理 $$\hat{L}_-Y_{lm}=b_{lm}Y_{l,m-1}$$

角动量算符

求: 常系数 $a_{l\,m}$, $b_{l\,m}$

$$\hat{L}_+ Y_{lm} = a_{lm} Y_{l,m+1}$$

$$\int (\hat{L}_+ Y_{lm})^* (\hat{L}_+ Y_{lm}) d\Omega = \int (a_{lm} Y_{l,m+1})^* (a_{lm} Y_{l,m+1}) d\Omega$$

$\int d\tau \psi^* \hat{O}^+ \varphi = \int d\tau (\hat{O}\psi)^* \varphi$

$\hat{L}_+^+ = \hat{L}_-$

$\hat{L}_- \hat{L}_+ = \hat{L}^2 - \hat{L}_z^2 - \hbar \hat{L}_z$

$\hat{L}_z Y_{lm} = m\hbar Y_{lm}, \hat{L}^2 Y_{lm} = l(l+1)\hbar^2 Y_{lm}$

$\int Y_{lm}^* Y_{l'm'} d\Omega = \delta_{ll'} \delta_{mm'}$

左 $= \int Y_{lm}{}^* \hat{L}_+{}^+ (\hat{L}_+ Y_{lm}) d\Omega$

$= \int Y_{lm}{}^* \hat{L}_- (\hat{L}_+ Y_{lm}) d\Omega$

$= \int Y_{lm}^* \left(\hat{L}^2 - \hat{L}_z^2 - \hbar \hat{L}_z \right) Y_{lm} d\Omega$

$= [l(l+1)\hbar^2 - m^2\hbar^2 - m\hbar^2] \int Y_{lm}^* Y_{lm} d\Omega$

$= l(l+1)\hbar^2 - m(m+1)\hbar^2$

右 $= |a_{lm}|^2 \int Y_{l,m+1}{}^* Y_{l,m+1} d\Omega = |a_{lm}|^2$

角动量算符

比较二式

$$|a_{lm}|^2=[l(l+1)-m(m+1)]\hbar^2$$

为简单计取实数：$a_{lm}=\hbar\sqrt{l(l+1)-m(m+1)}$

同理求得：$b_{lm}=\hbar\sqrt{l(l+1)-m(m-1)}$

$$\hat{L}_{\pm}Y_{lm}=\hbar\sqrt{l(l+1)-m(m\pm1)}Y_{l,m\pm1}=\hbar\sqrt{(l\mp m)(l\pm m+1)}Y_{l,m\pm1}$$

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证明

1、$\hat{L}_-\hat{L}_+ = \hat{L}^2 - \hat{L}_z^2 - \hbar\hat{L}_z$

2、$b_{lm} = \hbar\sqrt{l(l+1) - m(m-1)}$

3、$\hat{L}_-^+ = \hat{L}_+$